

Review for Quiz 5

Math 2270

Ex 1 Find an ON basis for $N(1, -1, 1, -1)$ Ex 2 Find an ON basis for $\left\langle \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ t \\ 1 \\ t \end{pmatrix} \right\rangle \quad t \neq 1$ Ex 3 Define ON basis of a vector space V Ex 4 If u is a unit vector, show that the columns of $Q = I - 2uu^T$ are an ON basisExplain why $x \mapsto Qx$ is a reflection about the hyperplane $u \cdot x = 0$ Ex 5 Compute $\det \begin{bmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{bmatrix}$

$$\det \begin{bmatrix} 3 & 3 & 4 \\ 6 & 8 & 7 \\ -3 & 5 & -9 \end{bmatrix}$$

$$\left(\begin{array}{l} \text{Answers:} \\ (b-a)(c-a)(c-b) \\ \& -6 \end{array} \right)$$

Ex 6 compute $\det(A - \lambda I)$ where $A = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 2 \\ 1 & 2 & 3 \end{pmatrix}$ $I = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$ Find the solutions of $\det(A - \lambda I) = 0$,and for these solutions $\bar{\lambda}$ compute $N(A - \bar{\lambda} I)$